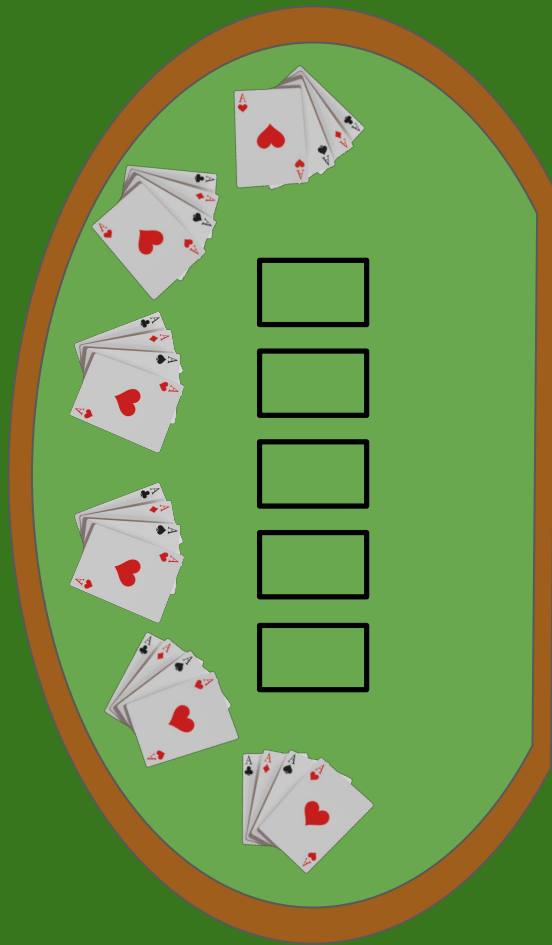




POKER FACE 2

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About Poker Face 2

What was Poker Face?

- One on one Texas Hold'em Poker playing AI
- Adaptive learning behavior
- Only reinforcement learning

What does Poker Face 2 add?

- Improved AI reasoning for poker game logic
- Multiplayer support for poker games



Poker Face 2 users and what they want

Target User Group

- People interested in poker without needing a full group of human players
- Players curious about AI behavior and strategy in competitive, imperfect-information games
- Computer science students and instructors interested in game AI and multi-agent systems

User Experience Summary

- Users can start a Texas Hold 'em game with any combination of human and AI players
- The interface presents a familiar poker table layout with clear betting, turn, and chip information
- AI opponents exhibit distinct playstyles and adapt to player behavior over time
- Gameplay is designed to be intuitive for poker players while still demonstrating advanced AI decision-making

How to play? + Current User Interface

- Texas Hold'em Poker Rules
- User Action Buttons
 - Check
 - Call
 - Raise (slider provided)
 - Fold
- Post-Game Actions
 - New Round
 - New Game
- AI settings
 - Reset AI Toggle training mode



Plan for Multiplayer Interface



Example UI From "World Series of Poker"

Justification

- The project aligns well with a senior capstone and my interest in game AI, using poker as a strategically rich domain to explore multi-agent AI through theory and implementation.
- It builds on my first-semester work by extending to multi-agent poker which will give me new experience creating multiplayer games.
- This six-player, online, multi-method agent approach will push me and challenge how much I can get done in a single semester (though I expect I will have to descope a bit to finish on schedule).

MVP vs Stretch goals

Minimum Viable Product (MVP)

- Fully playable Texas Hold 'em poker game supporting three players
- Ability to mix human and AI players in any combination
- At least one trained AI agent capable of making valid betting, folding, and bluffing decisions
- Functional UI that clearly displays game state, bets, and turns
- Stable game loop with correct poker rules and hand evaluation

Stretch Features

- Fully playable Texas Hold 'em poker game supporting six players
- Multiple AI agents trained with different learning or algorithms
- AI opponents with distinct, identifiable playstyles (e.g., aggressive, conservative, adaptive)
- Enhanced opponent modeling that adapts to multiple players simultaneously
- Improved UI elements such as animations, visual polish, or configurable difficulty levels

Technologies

Python

- Primary language for the project
- Easy and familiar

Pytorch

- Useful tool for designing AI agents
- Compatible with python

Pygame

- Previous UI was built in pygame since it works with python
- May explore other UI options that offer better features



Constraints and Assumptions

- The project will be developed by a single student within one semester
- Training will be limited by available local hardware and time constraints
- The UI will prioritize functionality and clarity over visual polish
- Poker rules will follow standard Texas Hold 'em rules
- Multiplayer will utilize online networking accessible via a web app

Questions?

